

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Coding Challenge

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 - Coding Challenge	Assessment:	ASSESSMENT TOOL 2 – Coding Challenge
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	To tackle coding challenge by preparing high efficiency algorithm		
Student Name:	M Poojitha Reddy	Reg. No.:	192472352
Section / Batch:	2024	Date:	26-08-26

Code of Conduct

I M Poojitha Reddy - 192472352 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M Poojitha Reddy

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. K-Means is sensitive to feature scaling. Using a dataset with features on very different scales (e.g., income in thousands vs. age in years), show how unscaled K-Means produces misleading clusters dominated by the high-magnitude feature, then fix it with standardization and demonstrate the corrected clustering.

CODE :

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

data = {
    'Age': [20, 22, 25, 28, 30, 35, 40, 42, 45, 50],
    'Income': [20, 22000, 25000, 28000, 30000, 55000, 60000, 62000, 90000, 95000]
}

df = pd.DataFrame(data)

X = df[['Age', 'Income']]

kmeans_unscaled = KMeans(n_clusters=3, random_state=42, n_init=10)
df['Unscaled_Cluster'] = kmeans_unscaled.fit_predict(X)

print("Clusters without scaling:")
print(df[['Age', 'Income', 'Unscaled_Cluster']])

plt.scatter(
    df['Age'],
    df['Income'],
    c=df['Unscaled_Cluster']
)
plt.xlabel('Age')
plt.ylabel('Income')
plt.title('K-Means Without Feature Scaling')
plt.show()

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
kmeans_scaled = KMeans(n_clusters=3, random_state=42, n_init=10)
df['Scaled_Cluster'] = kmeans_scaled.fit_predict(X_scaled)

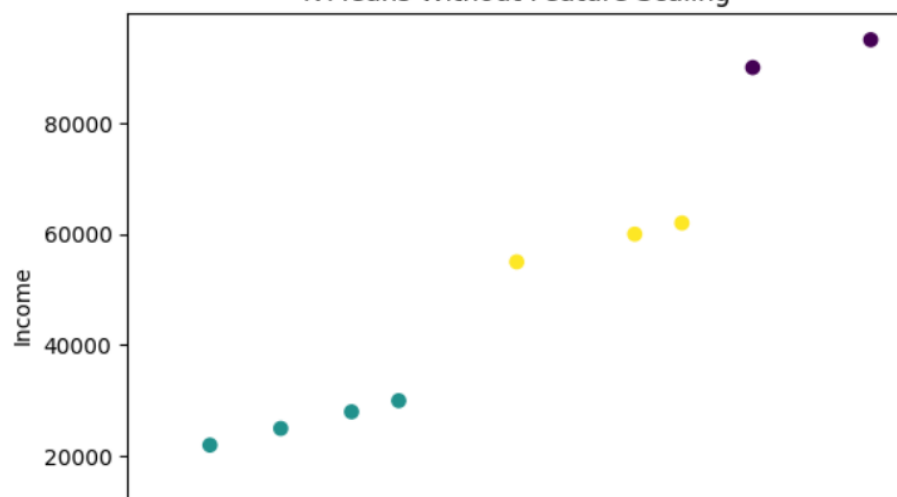
print("\nClusters after standardization:")
print(df[['Age', 'Income', 'Scaled_Cluster']])

plt.scatter(
    df['Age'],
    df['Income'],
    c=df['Scaled_Cluster']
)
plt.xlabel('Age')
plt.ylabel('Income')
plt.title('K-Means After Standardization')
plt.show()
```

OUTPUT :

```
Clusters without scaling:
***   Age  Income  Unscaled_Cluster
0    20     20           1
1    22    22000           1
2    25    25000           1
3    28    28000           1
4    30    30000           1
5    35    55000           2
6    40    60000           2
7    42    62000           2
8    45    90000           0
9    50    95000           0
```

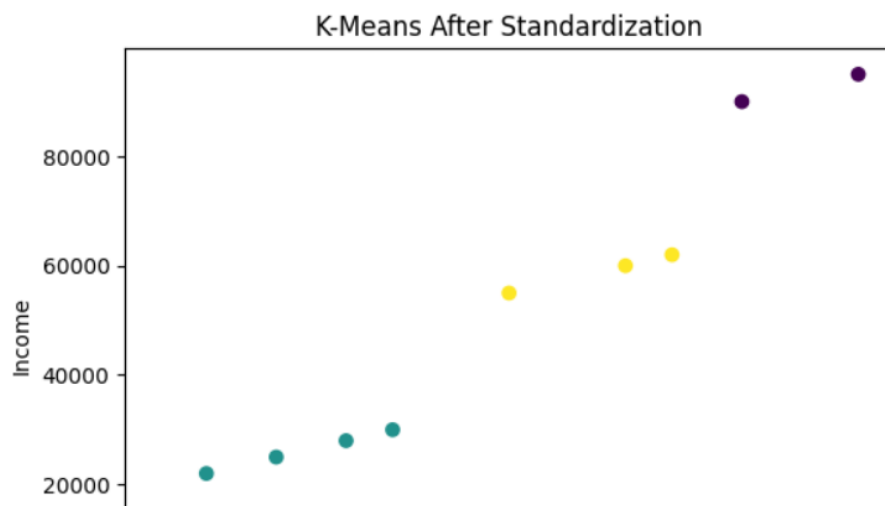
K-Means Without Feature Scaling



s  Terminal



```
Clusters after standardization:
...   Age  Income  Scaled_Cluster
0    20    20      1
1    22  22000    1
2    25  25000    1
3    28  28000    1
4    30  30000    1
5    35  55000    2
6    40  60000    2
7    42  62000    2
8    45  90000    0
9    50  95000    0
```



Terminal



GEOTAG PICTURE :

